

Report: Data Mining Project task 2

Introduction:

In the project, the two data sets namely iris dataset and breast cancer dataset are used to compare the accuracy, training time and prediction time of the defined KNN classifier, already defined KNN classifier from the module Sci-learn and Support vector Classifiers from the same module.

Experimental Setup:

The iris data set contains the 150 samples of flowers of 3 types based on the 4 attributes which are all numeric.

The another data set taken into account called breast cancer contain 569 samples of cancer cells and their 30 different numerical attributes. Based on those attributes we have to classify whether if the given cancer is malignant or benign.

The following classifiers were evaluated:

1. Custom kNN Classifier(Defined in the program) :

A k-Nearest Neighbors classifier implemented from scratch in Python. Euclidean distance was used as the distance metric, and class assignment was based on majority voting among the k nearest neighbors.

2. Scikit-learn kNN Classifier :

The KNeighborsClassifier from scikit-learn was used to compare the performance and runtime of training a classifier and testing it.

3. Support Vector Machine (SVM) :

A Support Vector Machine classifier with a radial basis function (RBF) kernel was used as a third comparison model.

For each dataset, the samples were split into training and test sets using a 70/30 split randomly. Stratified sampling was applied to ensure that class distributions in the training and test sets matched those of the original datasets. In the program, producing the testing samples randomly from the same data sets were taken care by train_test_split function in the sci-learn.

The code was executed two times using different testing and training samples chosen randomly from the datasets.

The required parameter for KNN classifier was chosen empirically based on the common suggestions from the literatures. For the SVM the default kernel function is used.

Presentation of the Results:

For accuracy scores were computed based on two runs for all three classifier the results are summarized in the table no. 1 below. With that the testing time (prediction time) and training time of all the classifier is also compared in the table no. 1.

Run No.	Data Set	Type of classifier	Accuracy	Training time(seconds)	Testing Time(seconds)
1	Iris	Custom Knn	0.955	4.52E-06	0.028
		KNN from Sci-Learn	0.955	0.00070	0.0015
		SVM	0.8666	0.00092	0.00028
	Breast-Cancer	Custom Knn	0.9122	2.86E-06	0.35
		KNN from sci-Learn	0.9122	0.00053	0.0012
		SVM	0.8713	0.0023	0.00117
2	Iris	Custom Knn	0.977	3.099E-06	0.02555
		KNN from Sci-Learn	0.9777	0.00057	0.0012
		SVM	0.9333	0.00078	0.00026
	Breast-Cancer	Custom Knn	0.92	1.19E-06	0.36
		KNN from Sci-Learn	0.92	0.00052	0.0012
		SVM	0.9122	0.0026	0.0010

Table no. 1

Discussion of results:

The custom KNN has much less training time than sci-learn KNN but sci-learn classifier beats custom one in testing time i.e. prediction time.

In comparison to the SVM the KNN always performs better on both runs because of the data being multi class. As expected , the training time of the SVM is more than both KNNs but on the other hand the prediction time is much smaller.

Conclusions :

In this study, a custom kNN classifier was implemented and evaluated alongside standard scikit-learn classifiers on benchmark datasets. The results show that the custom implementation performs competitively in terms of accuracy, while differences in runtime reflect the impact of algorithmic optimization.

Overall, the experiment shows that the KNN performs slightly better in terms of the accuracy than SVM with tradeoff between the prediction time and training time between the two.

References :

- Fisher, R. (1936). Iris [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C56C76>.
- Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5DW2B>.